

## **DETAILED DESCRIPTION OF THE INVENTION**

### **Mechanical Reference Frame and Geometric Constraint Definitions**

In various embodiments, the penile anatomical geometry profiling system operates within a defined mechanical reference frame to ensure spatial consistency during radial distance acquisition and longitudinal positional indexing. The reference frame may include a primary longitudinal axis corresponding to the anatomical centerline and one or more radial reference planes oriented relative to the housing structure of the sensing device.

The device may define a fixed internal reference plane from which radial distance measurements are calculated. This reference plane may correspond to a sensor mounting surface, an internal calibration datum, or a geometrically defined structural boundary within the housing assembly. By maintaining a consistent internal reference plane, radial measurements remain standardized across sessions and across devices manufactured within tolerance specifications.

In certain embodiments, geometric constraint modeling may be applied during reconstruction to enforce physically plausible anatomical continuity. Constraint conditions may include smoothness continuity along the longitudinal axis, bounded curvature thresholds, non-negative radial values, and continuity between adjacent cross-sectional reconstructions. Such constraints may reduce reconstruction artifacts arising from sampling noise or transient signal interference.

The system may further define a start-point reference corresponding to a reproducible anatomical landmark. Measurement routines may require confirmation of baseline alignment prior to data acquisition. Longitudinal displacement values may be indexed relative to this defined origin to preserve reproducibility between sessions.

In embodiments utilizing multi-sensor ToF arrays positioned at angular offsets, each sensor may be mapped into a common coordinate system prior to surface reconstruction. Transformation matrices or rotational alignment coefficients may be applied to convert sensor-specific distance values into unified radial coordinates relative to the primary longitudinal axis.

The geometric reference frame definitions described herein are not limited to any specific coordinate system, origin selection method, or constraint modeling approach. Any structured framework capable of correlating synchronized radial distance measurements with indexed longitudinal displacement values to generate a spatially coherent penile anatomical geometry profile falls within the scope of the present disclosure.